

Climate-Informed Trading Signals in the Global Wheat Market: A Multi-Factor, Algorithmic Approach

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Abstract

This paper investigates whether climate and environmental information can be systematically transformed into profitable trading signals in the global wheat market. We link daily and sub-seasonal temperature anomalies, precipitation deviations, drought indices, and satellite-derived vegetation indicators to returns on wheat futures (CME ZW) and wheat-focused exchange-traded funds (ETFs) such as WEAT and DBA. Building on the agronomic literature on temperature-yield relationships and remote-sensing-based yield forecasting, and on commodity futures as an asset class, we construct a multi-factor framework that combines: (i) linear and non-linear regression models, (ii) time-series models with exogenous climate factors, and (iii) machine learning algorithms (tree-based and sequence models). We design and backtest climate-driven trading strategies, with explicit treatment of transaction costs, slippage, risk limits, and position sizing. The paper provides a full blueprint—including model specification, feature engineering, algorithmic structure, and risk management design—to evaluate whether climate anomalies create persistent, tradeable inefficiencies in wheat pricing.

1 Introduction

Wheat is a fundamental agricultural commodity whose pricing is directly shaped by real environmental and biological processes. Unlike equities or macro instruments, wheat exhibits price sensitivity not only to macro supply and demand but also to micro-scale climate anomalies: temperature shocks, frost events, drought, excessive rainfall, and phenological stress at critical growth stages. As financialization increases, instruments such as CME wheat futures and wheat-focused ETFs (WEAT, DBA) have become liquid enough for systematic trading.

This paper evaluates whether climate and environmental data can be transformed into actionable trading signals in the wheat market. We build a multi-factor framework incorporating meteorological data, remote sensing vegetation indices, market microstructure variables, and macro controls. We then deploy linear, non-linear, time-series, and machine learning models to estimate

return predictability and to generate implementable signals for futures and ETFs. Finally, we design complete systematic strategies and evaluate their performance using realistic slippage, bid–ask spreads, volatility scaling, and execution cost models.

Our contribution is both methodological and practical: we provide a full, quantitative, reproducible pipeline for converting climate data into trade-ready signals within a professional algorithmic-trading research framework.

2 Literature Review

2.1 Climate Impacts on Wheat Yields

Empirical agronomy establishes that wheat yields are driven by: (1) cumulative growing degree days (GDD), (2) extreme heat days above physiological thresholds, (3) frost days during early growth and heading stages, and (4) water stress resulting from drought or insufficient precipitation.

Studies such as ? and ? provide quantitative mappings from weather to yields using panel regressions and threshold-based models. These yield sensitivities motivate our selection of climate variables (temperature anomalies, frost/heat counts, rainfall deviations, drought indices).

2.2 Remote Sensing and Vegetation Indices

Remote sensing (MODIS, Landsat, Sentinel) offers high-frequency, spatially-granular vegetation signals. Indices such as NDVI, SAVI, and DVI correlate strongly with biomass and yield potential. Research from NASA and others (???) demonstrates that vegetation signals captured during peak canopy formation can forecast wheat yields well before harvest.

These indices enter our model as medium-horizon predictors of supply-side tightening or loosening.

2.3 Commodity Pricing and Risk Premia

The literature on commodity futures (e.g., ?, ?) positions commodities as a return-generating asset class with roll yield and term-structure dynamics. Wheat futures respond to storage costs, inventory levels, and expectations of future supply—an endogenous channel through which climate-driven yield expectations affect prices.

The integration of agronomic shocks into financial returns remains underexplored, providing a strong motivation for this work.

3 Market Background

3.1 Wheat Production and Trade Flows

Wheat production is geographically concentrated across:

- U.S. Northern Plains (spring wheat)
- U.S. Southern Plains (winter wheat)
- Russia and Ukraine (Black Sea exporters)
- European Union
- Australia (export-oriented)

Climate anomalies in any of these regions can alter global supply expectations, thereby impacting futures prices.

3.2 Financial Instruments

CME Wheat Futures (ZW). Highly liquid; roll yield depends on curve slope (contango/backwardation); seasonality around planting and harvest.

WEAT ETF. Tracks wheat via futures exposure; subject to roll drag and tracking error.

DBA ETF. Broad agricultural exposure; useful for robustness tests.

4 Data

4.1 Market Data

We gather:

- ZW nearby and deferred futures prices
- ETF prices: WEAT, DBA
- Volume, open interest
- Volatility estimates: realized, GARCH

Continuous futures are constructed using:

$$r_{t+1} = \ln \left(\frac{P_{t+1}}{P_t} \right). \quad (1)$$

Roll yield:

$$\text{RY}_t = \frac{F_{t,m} - F_{t,n}}{F_{t,m}}. \quad (2)$$

4.2 Climate Data

Regions $j \in \mathcal{J}$ include: U.S. Plains, Ukraine, Russia, Australia.

We collect:

- Temperature (min, max, mean)
- Temperature anomalies
- GDD accumulation
- Frost and heat extreme counts
- Precipitation anomalies
- Drought indices (PDSI, SPI)
- Soil moisture

Anomalies:

$$T_{t,j}^{\text{anom}} = \frac{T_{t,j} - \bar{T}_j}{\sigma_{T,j}}. \quad (3)$$

GDD:

$$\text{GDD}_{t,j} = \sum_{s \leq t} \max(0, T_{s,j} - T_{\text{base}}). \quad (4)$$

Frost days:

$$\text{Frost}_{t,j} = \sum_{s \leq t} \mathbf{1}\{T_{s,j}^{\min} \leq \theta_f\}. \quad (5)$$

4.3 Remote Sensing Data

NDVI peak:

$$\text{NDVI}_j^{\text{peak}} = \max_{t \in \mathcal{W}_j} \text{NDVI}_{t,j}. \quad (6)$$

4.4 Macro and Geopolitical Controls

These include DXY, WTI crude oil, natural gas, and freight indices.

5 Methodology

5.1 Factor Construction

Define feature vector:

$$\mathbf{x}_t = \left[\mathbf{x}_t^{wx}, \mathbf{x}_t^{veg}, \mathbf{x}_t^{mkt}, \mathbf{x}_t^{macro} \right].$$

Return forecasting task:

$$\hat{r}_{t+1} = f(\mathbf{x}_t).$$

We consider multiple modeling families.

5.2 Linear Regression Models

Baseline:

$$r_{t+1} = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t + \varepsilon_t.$$

Distributed lags:

$$r_{t+1} = \alpha + \sum_{\ell=0}^L \boldsymbol{\beta}_\ell^\top \mathbf{x}_{t-\ell} + u_t.$$

Panel model:

$$r_{i,t+1} = \alpha_i + \boldsymbol{\beta}^\top \mathbf{x}_{i,t} + \epsilon_{i,t}.$$

5.3 Time-Series Models

ARIMAX:

$$\phi(B)(1-B)^d r_t = \theta(B)\epsilon_t + \boldsymbol{\gamma}^\top \mathbf{x}_t.$$

GARCH variance:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

VAR with exogenous climate factors:

$$\mathbf{y}_t = \sum_{k=1}^p A_k \mathbf{y}_{t-k} + B \mathbf{x}_t + \boldsymbol{\epsilon}_t.$$

5.4 Machine Learning Models

Tree-Based Models. XGBoost, Random Forest.

Sequence Models. LSTM using climate and vegetation sequences over a window L .

6 Signal Construction

Signal:

$$s_t = \frac{\hat{r}_{t+1}}{\hat{\sigma}_{t+1}}.$$

Position with risk targeting:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_{t+1}} \cdot s_t.$$

With cap:

$$|w_t| \leq w_{\max}.$$

7 Backtesting Framework

Walk-forward:

1. Train on rolling window
2. Forecast returns and volatility
3. Generate signals
4. Apply execution model (Appendix B)
5. Compute P&L
6. Update statistics and risk metrics

Transaction cost model integrated via:

$$\text{PnL}_{t+1} = w_t(P_{t+1} - P_t) - \text{TC}_{t+1}.$$

8 Results (Structure Only)

8.1 Descriptive Statistics

Present:

- Climate variable summaries
- Correlations
- Seasonal patterns

8.2 Model Performance

- In/out-of-sample errors
- Feature importance
- Region-level relevance

8.3 Strategy Performance

Metrics:

- CAGR, Sharpe, Sortino
- Max drawdown
- Turnover
- Regime-level breakdown

9 Discussion

We interpret:

- Predictive power of climate signals
- Strength of non-linear interactions
- Cross-region spillovers
- Macro interactions

Limitations discussed: data quality, shifts in crop management, liquidity constraints.

10 Robustness

Tests:

- Subsamples
- Region exclusion
- Extreme-event stress
- Parameter perturbation

11 Conclusion

We provide a full, deployable pathway for transforming climate information into tradeable signals for wheat futures and ETFs. The framework integrates climate science, remote sensing, econometrics, and modern machine learning alongside realistic execution cost modeling.

This pipeline forms a robust foundation for Hephaestus-style algorithmic research on climate-based commodities trading.

Disclaimer: This research is for academic and educational purposes only.

Appendix A: Hyperparameter Grids, Model Architectures, and Statistical Assumptions

This appendix provides full technical specifications for all model classes used in the paper. For completeness, we include optimization constraints, parameter bounds, learning schedules, and statistical assumptions underlying the models.

A.1 XGBoost Regressor (Full Grid)

We consider the following hyperparameter grid:

$$\begin{aligned}\eta &\in \{0.01, 0.05, 0.1, 0.2\}, \\ \text{max_depth} &\in \{3, 4, 5, 6, 8, 10\}, \\ \text{min_child_weight} &\in \{1, 5, 10, 50\}, \\ \text{gamma} &\in \{0, 0.1, 0.25, 0.5\}, \\ \text{subsample} &\in \{0.5, 0.7, 0.9, 1.0\}, \\ \text{colsample_bytree} &\in \{0.3, 0.5, 0.7, 1.0\}, \\ \text{lambda, alpha} &\in \{0, 0.1, 1, 10, 100\}, \\ \text{n_estimators} &\in \{300, 500, 800, 1200\}.\end{aligned}$$

XGBoost solves:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N \ell(r_i, f(\mathbf{x}_i)) + \sum_{k=1}^K \Omega(f_k),$$

where

$$\Omega(f_k) = \gamma T_k + \frac{\lambda}{2} \|w_k\|_2^2.$$

A.2 Random Forest Regressor Grid

$$\text{n_trees} \in \{500, 1000, 1500\}, \quad \text{max_features} \in \{\sqrt{d}, d/3, d/2\}, \quad \text{max_depth} \in \{6, 10, 20, 50, \infty\}.$$

Each tree is trained on a bootstrap sample:

$$\mathcal{D}_b^* = \{(\mathbf{x}_{i_b}, r_{i_b})\}_{b=1}^N,$$

with randomness injected via feature subsampling.

A.3 LSTM Architecture (Full Specification)

Given climate sequences:

$$\mathbf{X}_t = (\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t) \in \mathbb{R}^{L \times d},$$

an LSTM computes:

$$h_t = \text{LSTM}(\mathbf{X}_t; \Theta), \quad \hat{r}_{t+1} = Wh_t + b.$$

Architecture:

- Lookback window: $L \in \{20, 30, 60, 90\}$.
- Hidden size: $H \in \{32, 64, 128, 256\}$.
- Layers: $\{1, 2, 3\}$.
- Dropout: $\{0.0, 0.1, 0.2, 0.5\}$.
- Optimizer: Adam with

$$\alpha \in \{10^{-5}, 3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon = 10^{-8}.$$

Loss:

$$\mathcal{L} = \frac{1}{N} \sum_t (r_{t+1} - \hat{r}_{t+1})^2 + \lambda \|\Theta\|_2^2.$$

A.4 ARIMAX Specification

General model:

$$\phi(B)(1 - B)^d r_t = \theta(B)\epsilon_t + \gamma^\top \mathbf{x}_t,$$

with backshift operator $Br_t = r_{t-1}$.

Grid:

$$p, q \in \{0, 1, 2, 3, 4, 5\}, \quad d \in \{0, 1, 2\}.$$

A.5 GARCH(1,1) and EGARCH

GARCH:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

EGARCH:

$$\log(\sigma_t^2) = \omega + \alpha \left| \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} + \beta \log(\sigma_{t-1}^2).$$

Bounds:

$$0 < \omega < 1, \quad 0 < \alpha < 1, \quad 0 < \beta < 1, \quad \alpha + \beta < 1.$$

Appendix B: Execution, Slippage, and Market Microstructure Models

This appendix formalizes the execution cost function used in backtesting.

B.1 Almgren–Chriss Optimal Execution

We model execution cost as:

$$C = \gamma x^2 + \eta \sum_{t=1}^T v_t^2,$$

where:

- x : total order size
- v_t : slice executed at interval t
- γ : permanent impact coefficient
- η : temporary impact coefficient

Price dynamics:

$$S_{t+1} = S_t + \sigma \epsilon_{t+1} - \gamma v_t.$$

Temporary impact:

$$\tilde{S}_t = S_t - \eta v_t.$$

Optimal solution for mean-variance objective:

$$v_t^* = \frac{\sinh(\kappa(T+1-t))}{\sinh(\kappa)} \frac{x}{\sum_{k=1}^T \sinh(\kappa k)}, \quad \kappa = \sqrt{\frac{\lambda \sigma^2}{\eta}}.$$

B.2 Kyle (1985) Linear Impact Model

Let q_t be the signed trade size. Price impact:

$$\Delta S_t = \lambda q_t + \sigma u_t,$$

where:

$$\lambda = \frac{\sigma}{\sqrt{\mathbb{E}[q_t^2]}}.$$

Expected cost:

$$\mathbb{E}[C] = \lambda q_t^2.$$

B.3 Volume Market-Share Model

For execution as fraction of market volume:

$$\text{TC}_t = s_t + I \left(\frac{|v_t|}{\text{ADV}_t} \right)^\alpha,$$

where:

$$\alpha \in [0.5, 1.0], \quad I = \text{impact coefficient}.$$

B.4 Slippage as a Function of Spread, Depth, Volatility

Effective cost model:

$$TC_t = \frac{\text{Spread}_t}{2} + \frac{\sigma_t}{\sqrt{D_t}} |v_t| + \chi v_t^2,$$

with order book depth D_t .

B.5 Execution Costs for ETFs (WEAT)

ETF execution must consider:

$$TC = \text{Spread}/2 + \text{CreationCost} + \text{Impact}.$$

Creation/redemption cost:

$$C_{\text{ETF}} = \text{BasketTrackingError} + \text{FuturesRollCost}.$$

For WEAT:

$$C_{\text{roll}} = \sum_{k \in \text{roll window}} (F_{k+1} - F_k) \cdot w_k.$$

B.6 Overall Execution Cost Function

All execution models are combined into:

$$TC_t = C_t^{AC} + C_t^{Kyle} + C_t^{MarketShare} + C_t^{Depth} + C_t^{ETF}.$$

Backtests assume:

$$\hat{TC}_t = \max(TC_t, \text{MinCost}).$$

Appendix C: Fully Specified Climate Trading Strategy

We present a complete strategy used in the paper: the Frost Shock Long Wheat strategy. This is designed as a canonical example of translating climatic signals into a deployable trading rule.

C.1 Climate Signal Definition

Define regional minimum temperature:

$$T_{t,j}^{\min}.$$

Define frost threshold (crop science literature):

$$\theta_f = -2^\circ\text{C}.$$

Define frost shock indicator:

$$F_{t,j} = \mathbf{1}\{T_{t,j}^{\min} \leq \theta_f\}.$$

Define multi-region frost stress:

$$F_t = \sum_{j \in \mathcal{J}} w_j F_{t,j}, \quad w_j = \text{regional acreage share.}$$

C.2 Expected Price Impact

We use a reduced-form local projection:

$$\mathbb{E}[r_{t+1}^{ZW} | F_t] = \alpha + \beta F_t.$$

Empirically $\beta > 0$.

We define:

$$\hat{r}_{t+1} = \hat{\alpha} + \hat{\beta} F_t.$$

C.3 Position Sizing with Volatility Targeting

Signal:

$$s_t = \frac{\hat{r}_{t+1}}{\hat{\sigma}_{t+1}}.$$

Vol-targeted position:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_{t+1}} \cdot s_t.$$

Cap:

$$w_t \leftarrow \min(\max(w_t, -w_{\max}), w_{\max}).$$

C.4 Execution and P&L

Actual filled price:

$$\tilde{P}_{t+1} = P_{t+1} + \text{TC}_{t+1}.$$

PnL:

$$\text{PnL}_{t+1} = w_t \cdot (P_{t+1} - P_t) - \text{TC}_{t+1}.$$

C.5 Strategy Pseudocode

Algorithm 1 Frost Shock Long ZW Strategy

- 1: **for** each day t **do**
 - 2: Compute frost indicator F_t .
 - 3: Forecast return: $\hat{r}_{t+1} = \hat{\alpha} + \hat{\beta}F_t$.
 - 4: Forecast volatility $\hat{\sigma}_{t+1}$ via GARCH.
 - 5: Compute signal $s_t = \hat{r}_{t+1}/\hat{\sigma}_{t+1}$.
 - 6: Set position $w_t = (\sigma_{target}/\hat{\sigma}_{t+1})s_t$.
 - 7: Cap position: $|w_t| \leq w_{\max}$.
 - 8: Execute trades using execution model (Appendix B).
 - 9: Compute P&L: $w_t(P_{t+1} - P_t) - TC_{t+1}$.
 - 10: Update rolling statistics and drawdown.
 - 11: **end for**
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References